

Data Analysis and Visualization of Developer Trends

Michal Bando



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OUTLINE



1. Executive Summary,
2. Introduction,
3. Methodology,
4. Language & Database Trends,
5. Dashboards,
6. Insights & Conclusions.

EXECUTIVE SUMMARY



- **Core Finding:** JavaScript remains the most utilized language, but Python and Rust show the highest growth in future interest.
- **Database Shift:** Traditional SQL databases (PostgreSQL/MySQL) dominate current usage, while NoSQL and Cloud-native solutions see rising future demand.
- **Demographics:** The workforce is primarily composed of individuals aged 25-34, highlighting a young, tech-savvy global demographic.



INTRODUCTION



- **Purpose:** To analyze and visualize current and future technology trends within the global developer community.
- **Target Audience:** CTOs, IT Recruitment Managers, and Software Architects.
- **Value:** Provides data-driven insights for strategic technology stack selection and talent acquisition planning.



METHODOLOGY



Data Sources: Stack Overflow Annual Developer Survey.

Collection Methods: Data was retrieved using **Web Scraping** (BeautifulSoup) and **REST APIs**.

Wrangling Steps:

Removed duplicate entries based on ResponseId.

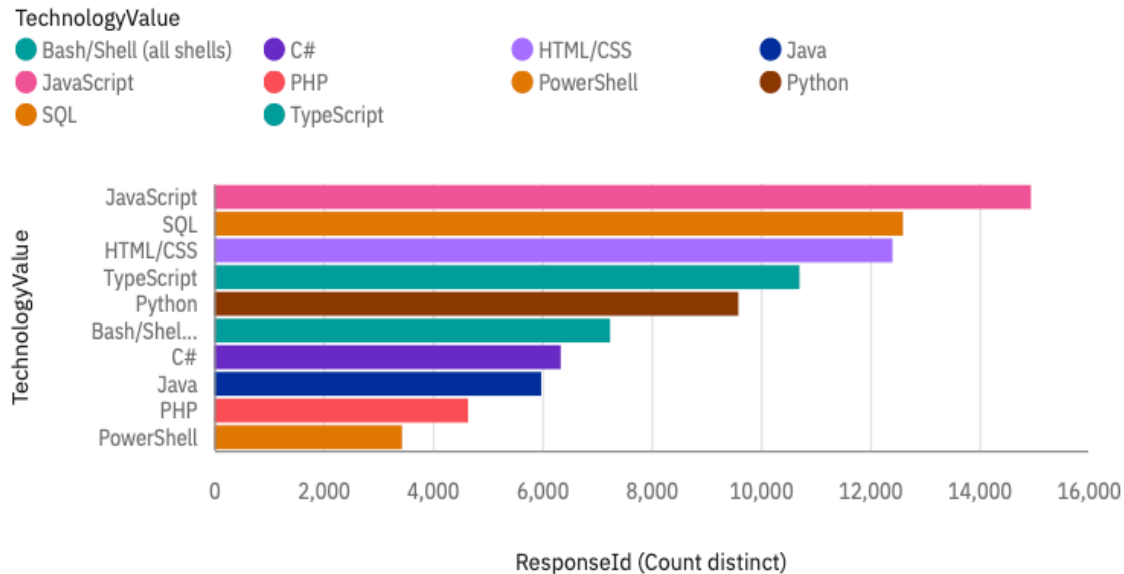
Handled missing values to ensure statistical integrity.

Normalization: Transformed multi-select columns (e.g., Languages) into a long format using Pandas explode for accurate frequency analysis.

PROGRAMMING LANGUAGE TRENDS

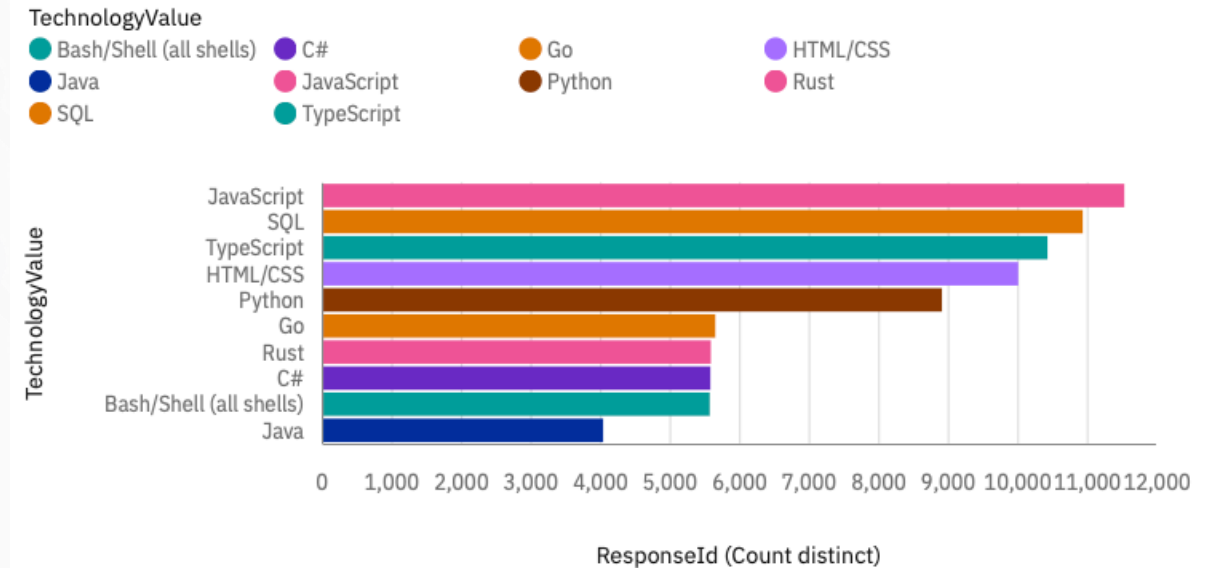
Current Year

Top 10 LanguageHaveWorkedWith



Next Year

Top 10 LanguageWantToWorkWith



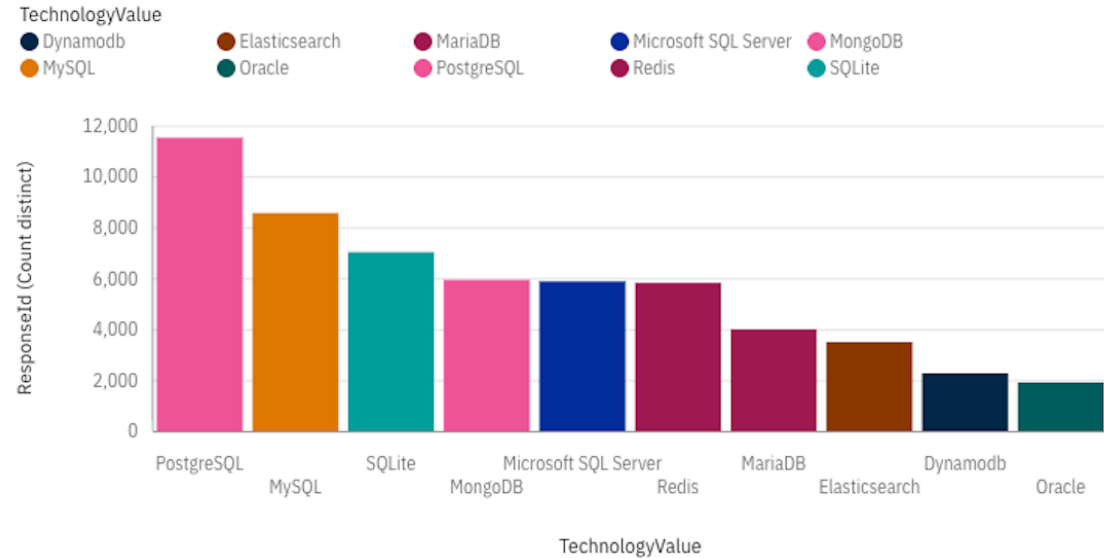
PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

- **Findings:** JavaScript continues to lead in current deployment, but Python is the most "Desired" language for the upcoming year.
- **Implications:** Organizations should focus on Python-related upskilling to align with developer interests and the rise of AI/Data Science.

DATABASE TRENDS

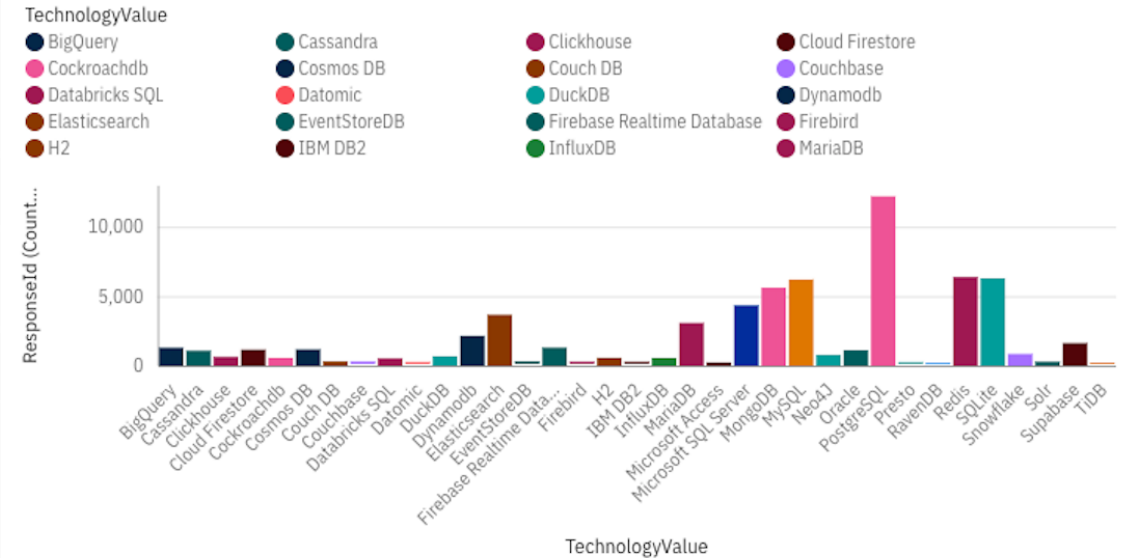
Current Year

Top 10 DatabaseHaveWorkedWith



Next Year

Top 10 DatabaseWantToWorkWith



DATABASE TRENDS - FINDINGS & IMPLICATIONS

- **Findings:** PostgreSQL and MySQL are the current industry standards, but there is significant growth in interest for MongoDB and Redis.
- **Implications:** The move toward NoSQL suggests a trend toward more flexible, scalable data architectures for modern applications.



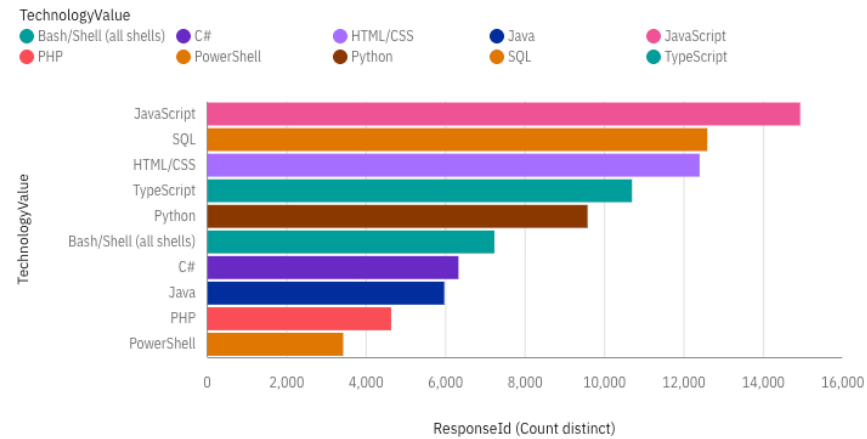
DASHBOARD



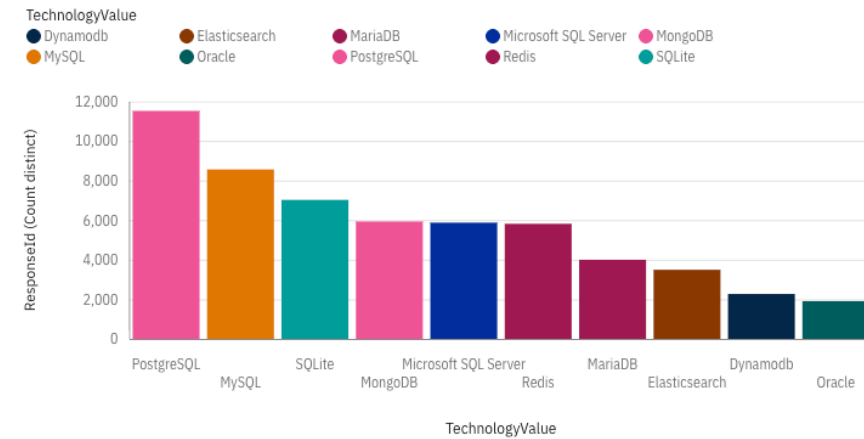
<Please present your dashboard in the following slides.>

DASHBOARD TAB 1

Top 10 LanguageHaveWorkedWith



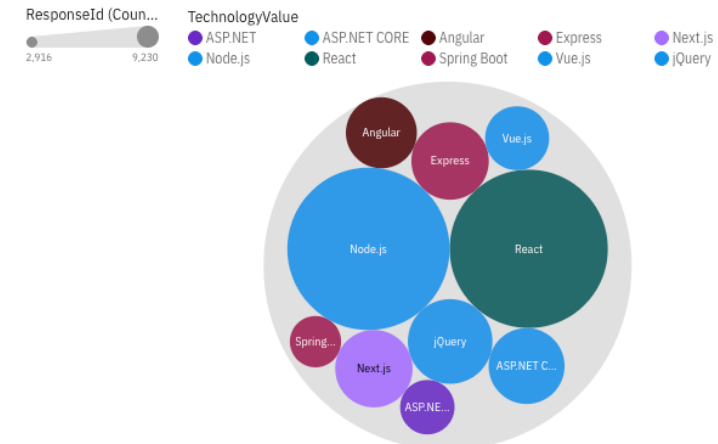
Top 10 DatabaseHaveWorkedWith



Top 10 PlatformHaveWorkedWith



Top 10 WebFrameHaveWorkedWith



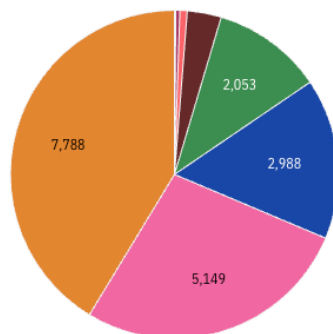


DASHBOARD TAB 3

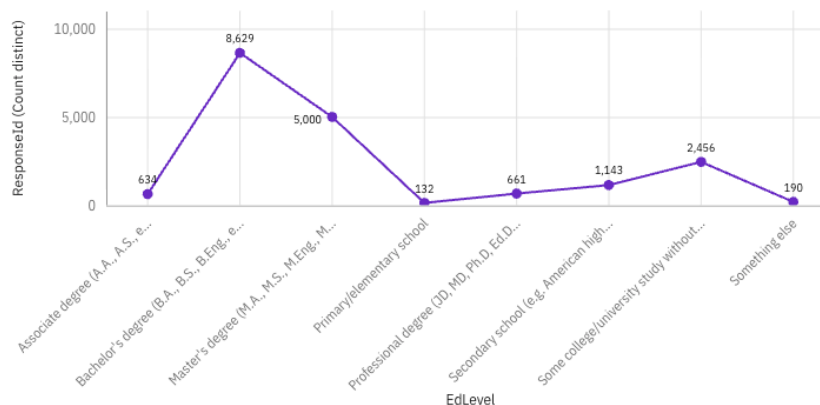
Distribution of Respondents by Age

Age

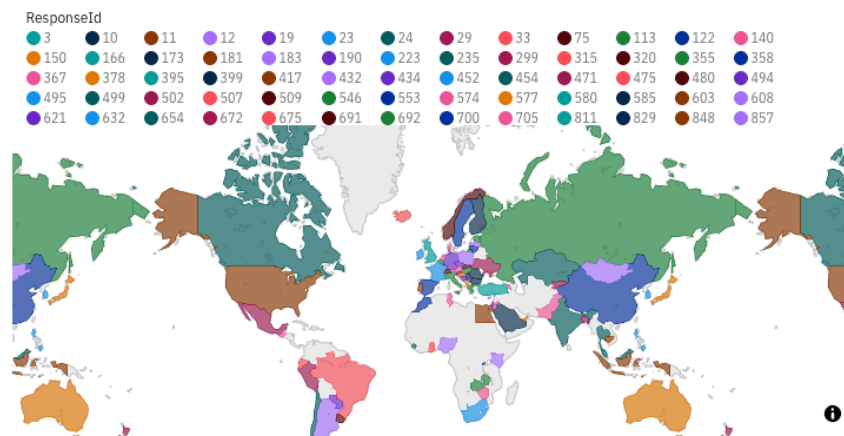
- Prefer not to say
- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65 years or older
- Under 18 years old



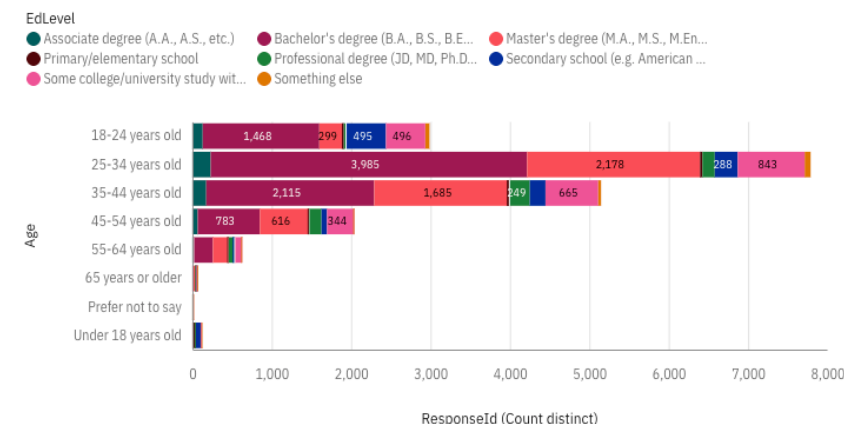
Distribution of Respondents by Education Level



Number of Respondents by Country



Number of Respondents by Age and Education Level



DISCUSSION



- **Tech Stack Evolution:** There is a clear transition from monolithic "standard" stacks toward diversified, specialized tools.
- **Demographic Alignment:** Most developers are early-to-mid career professionals, indicating a high potential for rapid adoption of new tools.
- **Geographic Concentration:** Major hubs in North America, Europe, and India continue to drive global technology standards.



OVERALL FINDINGS & IMPLICATIONS

- **Skill Gap:** The gap between "Worked With" and "Want to Work With" suggests an upcoming shift in the talent market toward AI and Cloud-native technologies.
- **Strategic Growth:** Companies leveraging the most "desired" technologies will likely have higher employee retention and recruitment success.



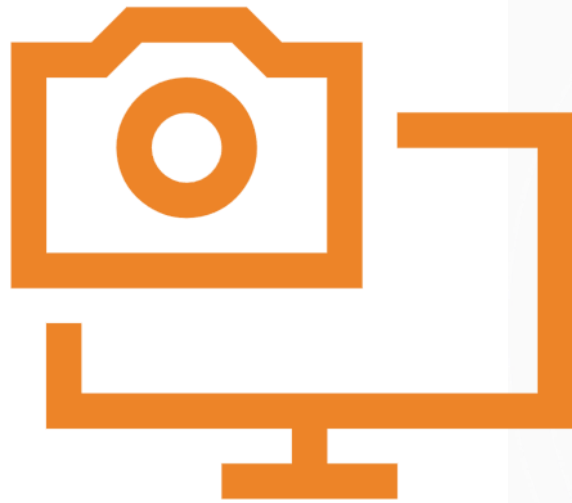
CONCLUSION



- JavaScript and SQL remain foundational to the current IT landscape.
- Python and NoSQL are the primary drivers of future technological interest.
- Effective data normalization and visualization are essential for translating raw survey data into actionable business intelligence.



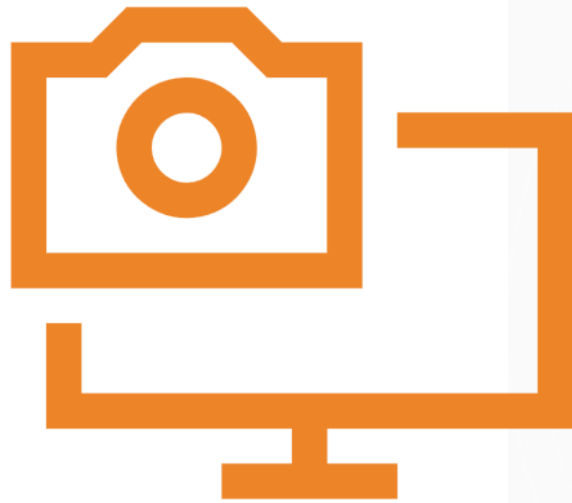
APPENDIX



- **Focus:** Explaining the technical "heavy lifting" done in Python.
- **The Challenge:** The raw Stack Overflow dataset stores multi-select responses (e.g., "Python;SQL;Java") as a single semi-colon-separated string. Standard BI tools cannot count these individual instances accurately.
- **The Solution:** Utilized the Python **Pandas library** and the `$explode()` function. This process split strings into individual rows, creating a "long-format" dataset.
- **The Impact:** This normalization ensures 100% accuracy in the "Top 10" charts. Without this step, unique technologies would be hidden within combined strings, leading to significant undercounting.



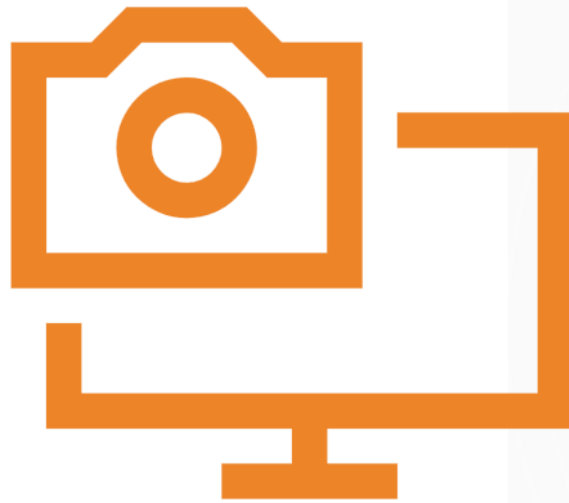
APPENDIX



- **Focus:** Demonstrating attention to data integrity.
- **Deduplication:** Performed rigorous data cleaning by removing duplicate records based on the unique \$ResponseId\$.
- **Handling Missing Values:** Implemented filtering to exclude null or "None" responses in technology columns to prevent "empty" categories from appearing in the dashboard.
- **Data Casting:** Converted string-based salary and age data into numerical types (floats/integers) to enable mathematical aggregations and "Count Distinct" measures.



APPENDIX



Text: "Implementation of the Normalization Pipeline using Python:

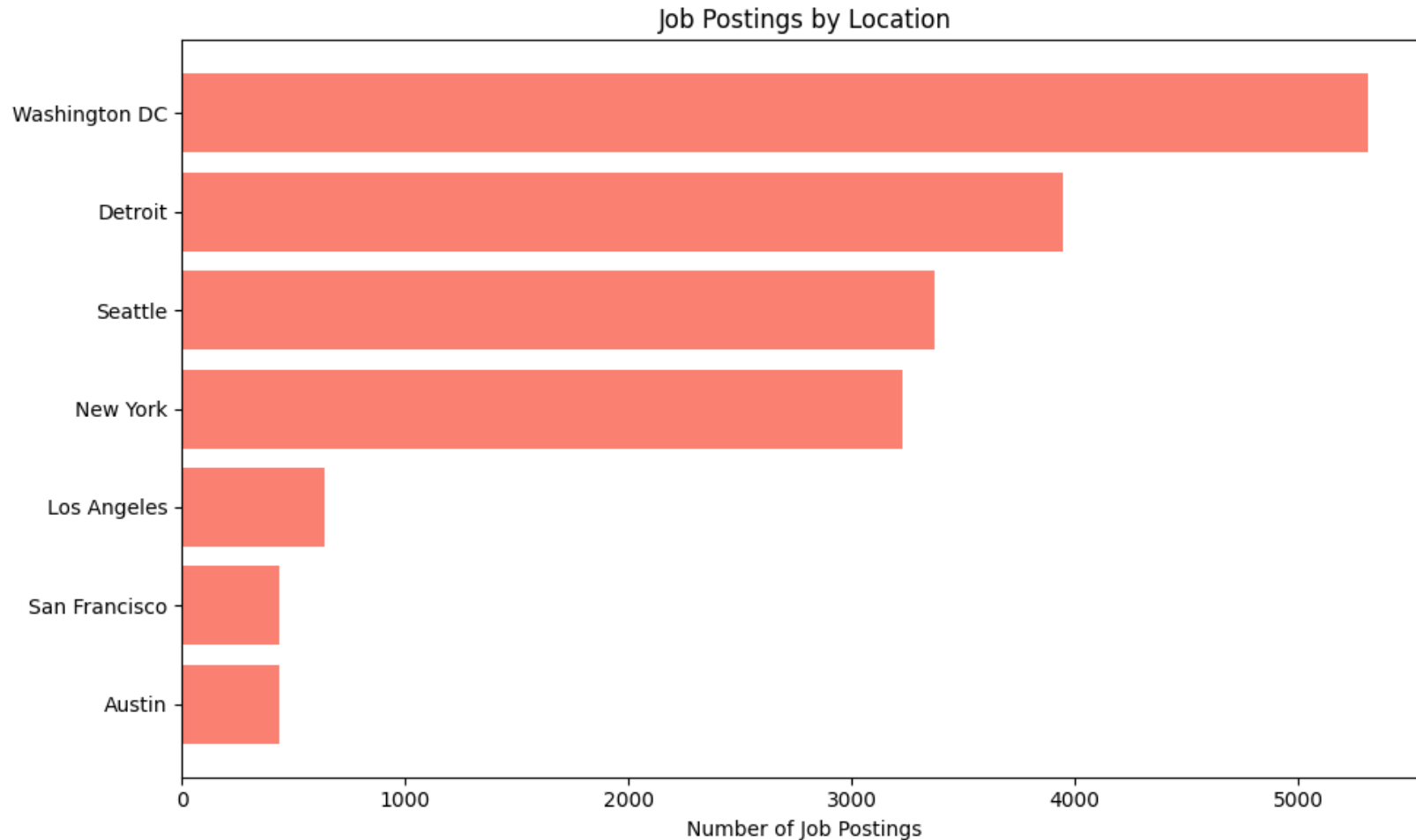
Splitting strings and exploding to long format

```
df['Languages'] = df['Languages'].str.split(';') df_final = df.explode('Languages')
```

- **Value-Add:** This snippet demonstrates the transition from raw data to a visualization-ready state.

JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.



POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.

